# KARTHIK CHANDRAN LAKSHMANASAMY

Alexandria, VA | **U.S. Permanent Resident | No Sponsorship Required**
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## PROFESSIONAL SUMMARY

Principal Software Engineer in Test with 9+ years of experience leading enterprise quality engineering across SaaS, cloud, distributed, data-intensive, and AI/LLM platforms. Expertise in Python/Pytest automation, UI/API testing, CI/CD, performance and reliability engineering, cloud-native testing, and risk-based test strategy. Proven track record modernizing legacy frameworks, validating RAG/LLM and semantic layer systems, building scalable automation across multi-cloud environments, and mentoring engineers on production-grade quality practices.

## CORE EXPERTISE

**AI/LLM & RAG Testing | Agentic AI & MCP Testing | Python / Pytest | Selenium / Playwright / WebdriverIO | API & Integration Testing | Performance & Concurrency Testing | CI/CD & Quality Guardrails | Docker / Kubernetes | AWS / Azure / GCP | Distributed Systems | Framework Architecture & Modernization**

## PROFESSIONAL EXPERIENCE

**Principal Software Engineer in Test** **May 2023 - Present**
*MicroStrategy* *Tysons, VA*

- Modernized a 20-year-old legacy automation framework by migrating core regression workflows to Python/Pytest and containerizing test execution and deployment with Docker/Kubernetes, improving maintainability, portability, and CI/CD readiness.
- Developed end-to-end automation frameworks using WebdriverIO and Python for Strategy Mosaic, improving regression coverage, test efficiency, and release confidence.
- Built a framework to validate Mosaic's Universal Semantic Layer architecture, including semantic models, metrics, relationships, query generation, SQL accuracy, and data consistency across enterprise data sources and BI platforms.
- Automated AI-based data modeling workflows, including linked models, base and derived metrics, automatic metric generation, and AI-powered analytics.
- Developed automated testing for Mosaic's Model Context Protocol (MCP) integration with Claude, validating connector setup, tool discovery, semantic model access, query execution, response accuracy, permissions, and production database or in-memory execution.
- Validated LLM-generated responses against Mosaic's governed semantic layer to ensure deterministic, traceable, grounded, and accurate business results.
- Implemented TensorFlow-based predictive validation for Retrieval-Augmented Generation (RAG) workflows, improving accuracy for forecast-related conversational queries.
- Built a Python automation framework from scratch to test customer-managed cloud deployments across AWS, Azure, GCP, OpenShift, and StackIT.
- Automated cloud provisioning, microservice health checks, configuration, upgrades, autoscaling, workload distribution, concurrency, recovery, and system stability under varying traffic volumes.
- Automated pre- and post-deployment microservice checks using AWS CodeBuild, validating AI services, SQL Engine functionality, and distributed system components.
- Designed Jenkins CI/CD workflows for automated functional, integration, deployment, and regression testing.
- Built a Python concurrency testing framework for event-driven data models to identify throughput, latency, scalability, and stability bottlenecks.
- Mentored junior engineers on automation architecture, cloud testing, and tooling selection, helping achieve 90% test coverage across critical components.

**Senior Software Engineer in Test** **Jul 2019 - May 2023**
*MicroStrategy* *Poland*

- Designed an automated framework using Protractor, JavaScript, and Cucumber, reducing manual testing effort for the SAP S/4 Connector, cloud-based applications, and services.
- Enhanced Kafka-Flink data stream performance by 38% by optimizing and validating batch size and processing intervals under data-intensive workloads.
- Played a key role in certification and release validation for ADLS, S3, and GCS cloud connectors in MicroStrategy 2021.
- Led FedRAMP certification testing for the Cloudera Kubernetes platform, ensuring compliance with federal security and platform requirements.
- Built a data pipeline to analyze server logs and surface performance metrics used to optimize backend services.

**Software Engineer in Test** **Mar 2018 - Jul 2019**

*MicroStrategy* *Tysons, VA*

- Designed a Pytest and Selenium framework to detect Google BigQuery data import issues, improving reliability of data connectivity workflows.
- Streamlined CI/CD workflows by partnering with DevOps to integrate automated tests into pipelines, improving deployment reliability and continuous release validation.
- Managed planning, integration, and continuous testing for the Google BigQuery connector in MicroStrategy.
- Led Teradata-to-Google BigQuery migration validation with cross-functional teams to improve migration quality and data accessibility.

**Associate Software Engineer in Test** **Mar 2017 - Mar 2018**

*MicroStrategy* *Tysons, VA*

- Designed a performance testing framework to benchmark big data connectors across large-scale workloads.
- Improved large-scale data import response time by 44% using the Google Storage API and performance-focused validation.
- Led continuous improvement initiatives to increase test efficiency, reduce flakiness, and improve defect detection.
- Collaborated with cross-functional teams to define, review, and prioritize requirements and features.
- Implemented and validated Kerberos authentication for Cloudera Data Platform using Active Directory, strengthening enterprise security.

## TECHNICAL SKILLS

**Programming:** Python, JavaScript, TypeScript, Java, Bash, Groovy, C++
**Test Automation:** Pytest, Selenium, Playwright, WebdriverIO, Cypress, Protractor, Cucumber, JUnit, TestNG
**AI / LLM Testing:** Generative AI, LLM evaluation, LLM-as-a-Judge, RAG validation, prompt/response grounding, TensorFlow, Azure Embeddings, LangChain, OpenAI, AWS Bedrock, Gemini, Claude, GPT-OSS, MCP
**API & Integration:** Postman, Swagger, REST/API validation, microservices, data connectors
**Performance & Reliability:** JMeter, Gatling, concurrency testing, load/performance benchmarking, scalability, regression testing
**CI/CD & DevOps:** Jenkins, AWS CodeBuild, Git, Docker, Kubernetes, Terraform, Ansible
**Cloud & Platforms:** AWS, Azure, GCP, OpenShift, StackIT
**Data & Distributed Systems:** Kafka, Flink, BigQuery, Databricks, Snowflake, Oracle, MySQL, MongoDB, PostgreSQL, Redshift, Redis
**BI / Enterprise:** MicroStrategy, Tableau, Qlik, Salesforce, ServiceNow, Power BI, ThoughtSpot, Amazon QuickSight

## EDUCATION

Master of Science in Computer Science | Texas State University | Aug 2013 - May 2016
Bachelor of Engineering in Computer Science | Anna University | Sep 2009 - May 2013